Digital-to-Analog Conversion

ECE 2564, Embedded Systems

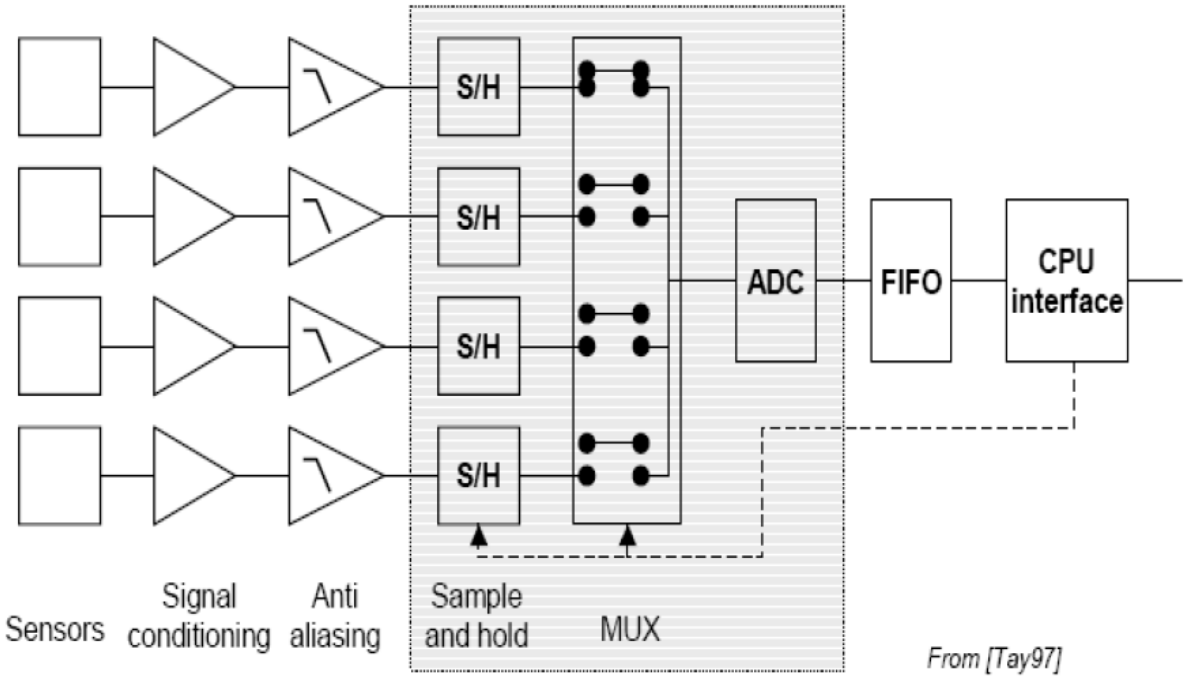
Fall 2025

Analog-to-Digital Conversion (Part 3)

- Signal Conditioning
- Sampling
- Additional discussion of ADC hardware

Analog-to-Digital Signal Conditioning





ADC requires signal conditioning

Signal Conditioning

- *Signal Conditioning* refers to the steps we take to prepare an analog signal for conversion to digital form
- Signals of interest are often *noisy* and so we need to consider methods for noise reduction
  - Possibly filter the input signal to remove high or low frequency noise
  - Maybe use an analog filter before the ADC
  - Maybe use a digital filter after the ADC if the signal is sampled are regular intervals
- Signals of interest often have amplitudes that don't match the input limits of an ADC
  - Weak signals need to be amplified
  - Large amplitude signals may need to be attenuated

Presence of Signal Noise

- Joystick example
  - Did you notice that the least significant bits (LSB) of the x and y positions were usually not stable?
- How to address the problem?
  - Design and implement an analog low-pass or band-pass filter
  - If the signal is sampled at regular intervals, after ADC take the average of the last k samples

Signal Amplification/Reduction

- EFSR: Effective full-scale range
- Dynamic Range (DR): the difference between a signal's maximum and minimum range

- An amplifier circuit can boost a weak signal, but it will also amplify the noise
- We typically want to amplify or reduce an input analog signal so that its *dynamic range* is close to the EFSR of the ADC
  - We want the DR to be close to the EFSR. Why? We dont' want the DR to be larger than the EFSR. Why?
  - Conditioning amplifier gain (G): We want DR $\times$ G $\leq$ EFSR

Example: Gain Conditioning

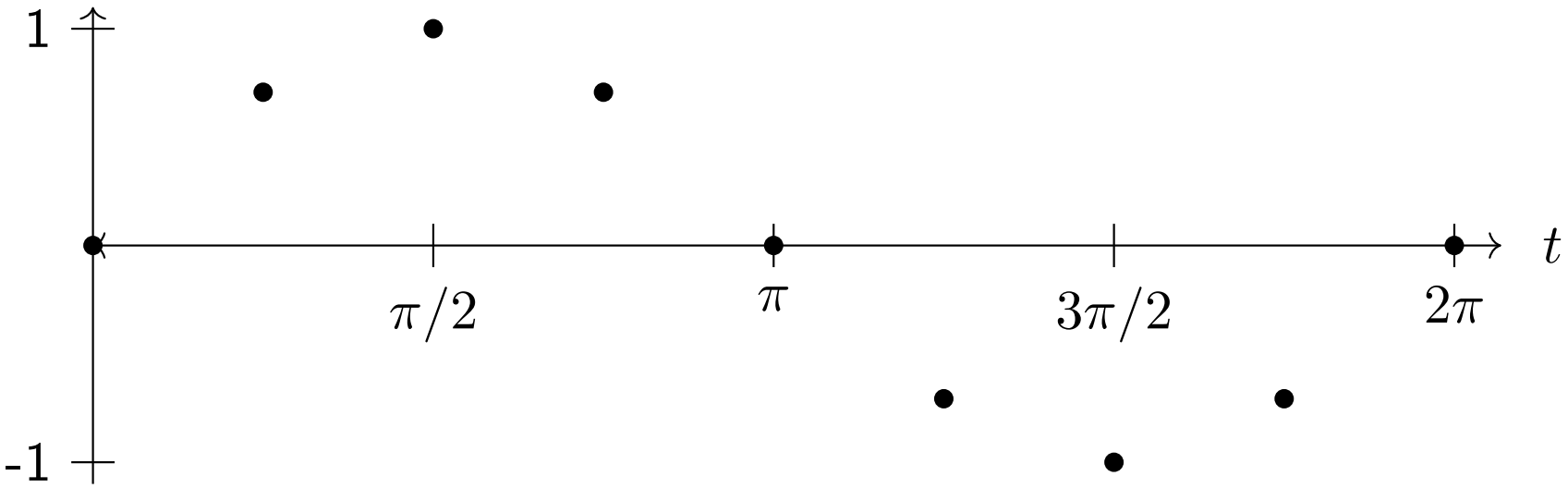
- Consider an input signal with a dynamic range of $20\,\text{mV}$
- Suppose EFSR $= 1\,\text{V}$

$$\begin{aligned} \text{DR} \times \text{G} &\leq \text{EFSR} \\ \text{G} &\leq \frac{\text{EFSR}}{\text{DR}} \\ &= \frac{1\,\text{V}}{20\,\text{mV}} \\ &= 50 \end{aligned}$$

Anti-Aliasing

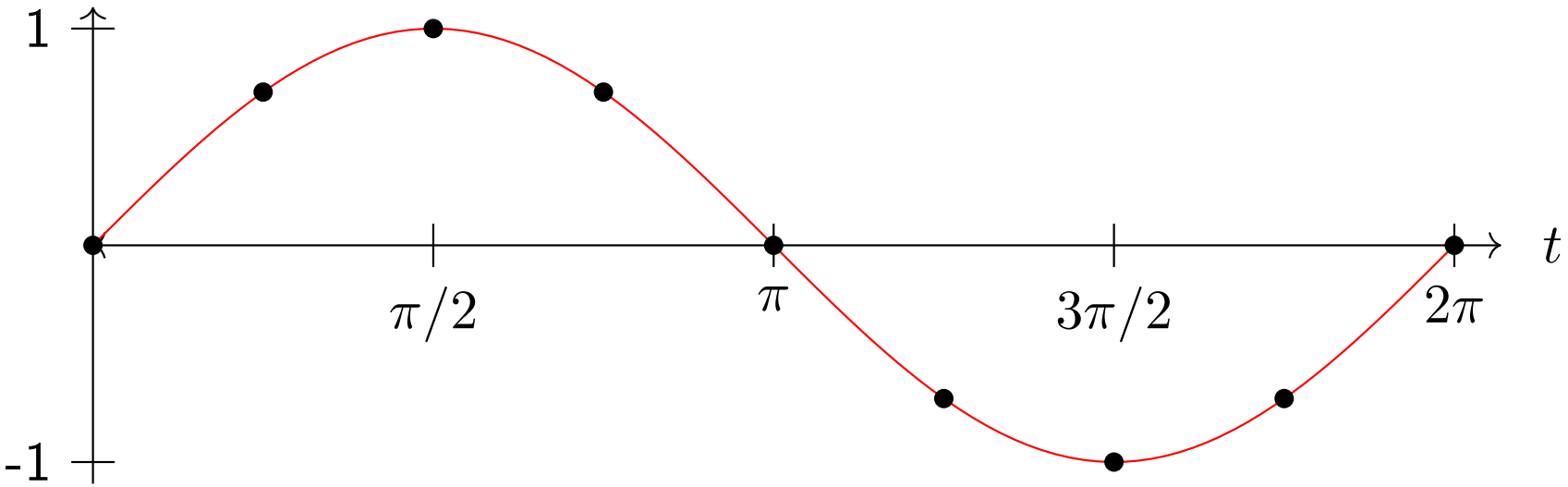
Example of aliasing

Aliasing Example: Sampled Signal



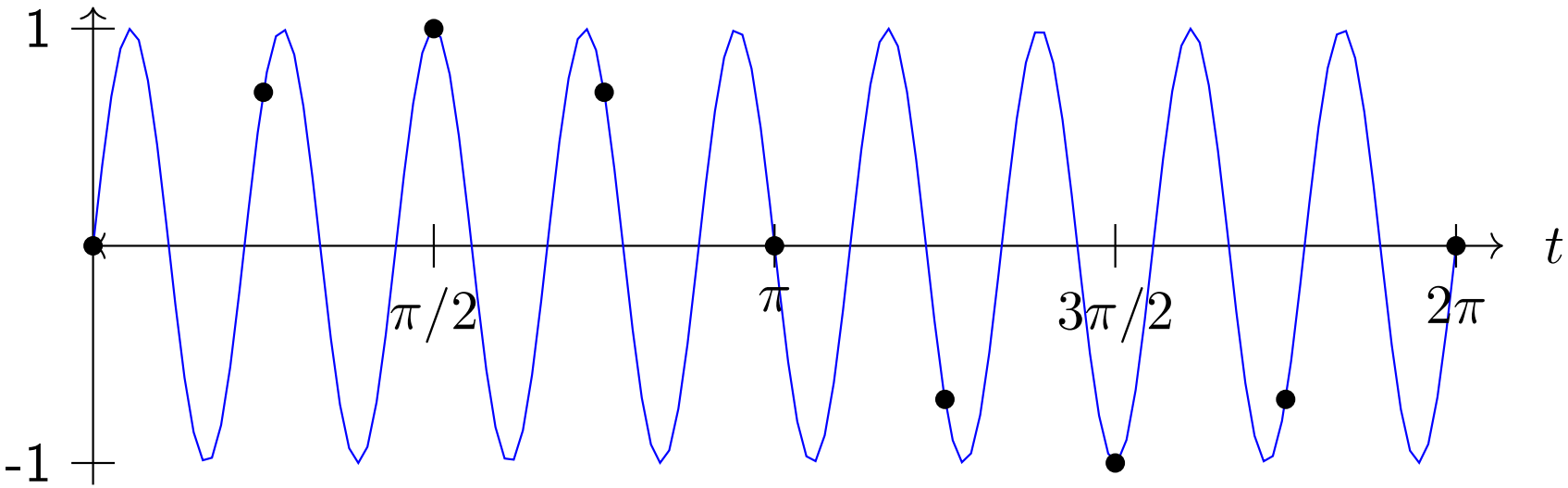
Aliasing Example: Sampled Signal Reconstruction

Did the sampled data come from $y(t) = \sin(t)$?



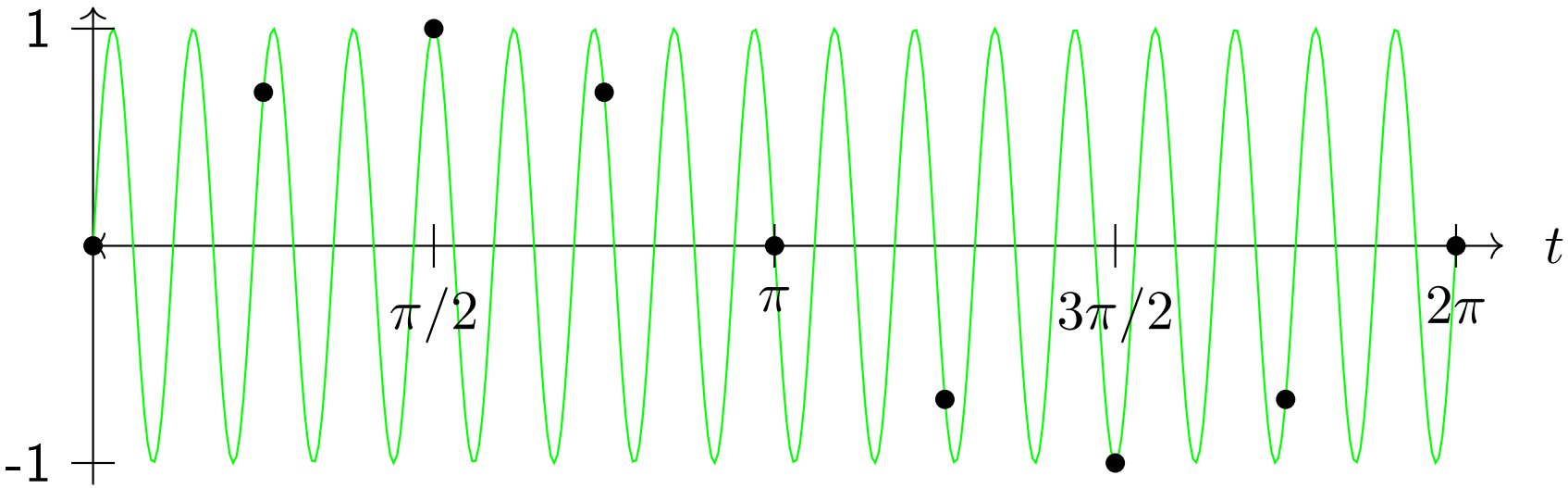
Aliasing Example: Sampled Signal Reconstruction

Did the sampled data come from $y(t) = \sin(9t)$?

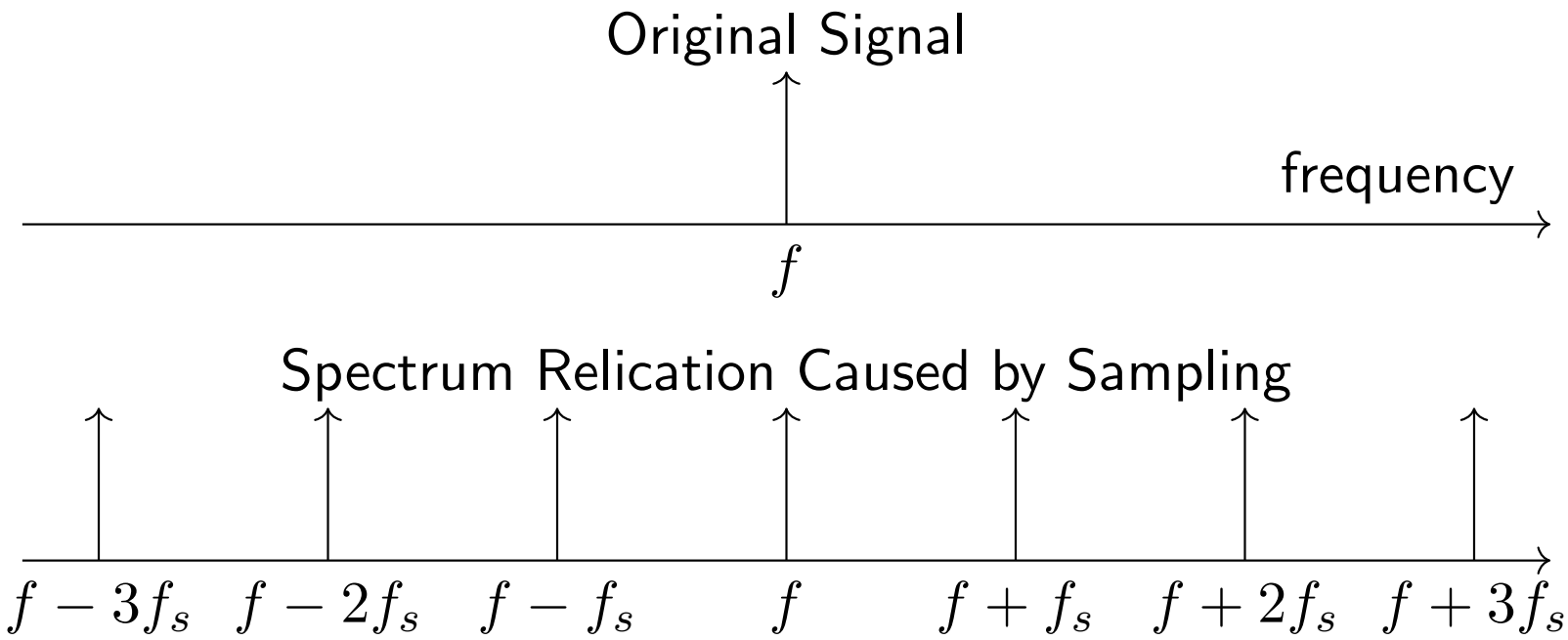


Aliasing Example: Sampled Signal Reconstruction

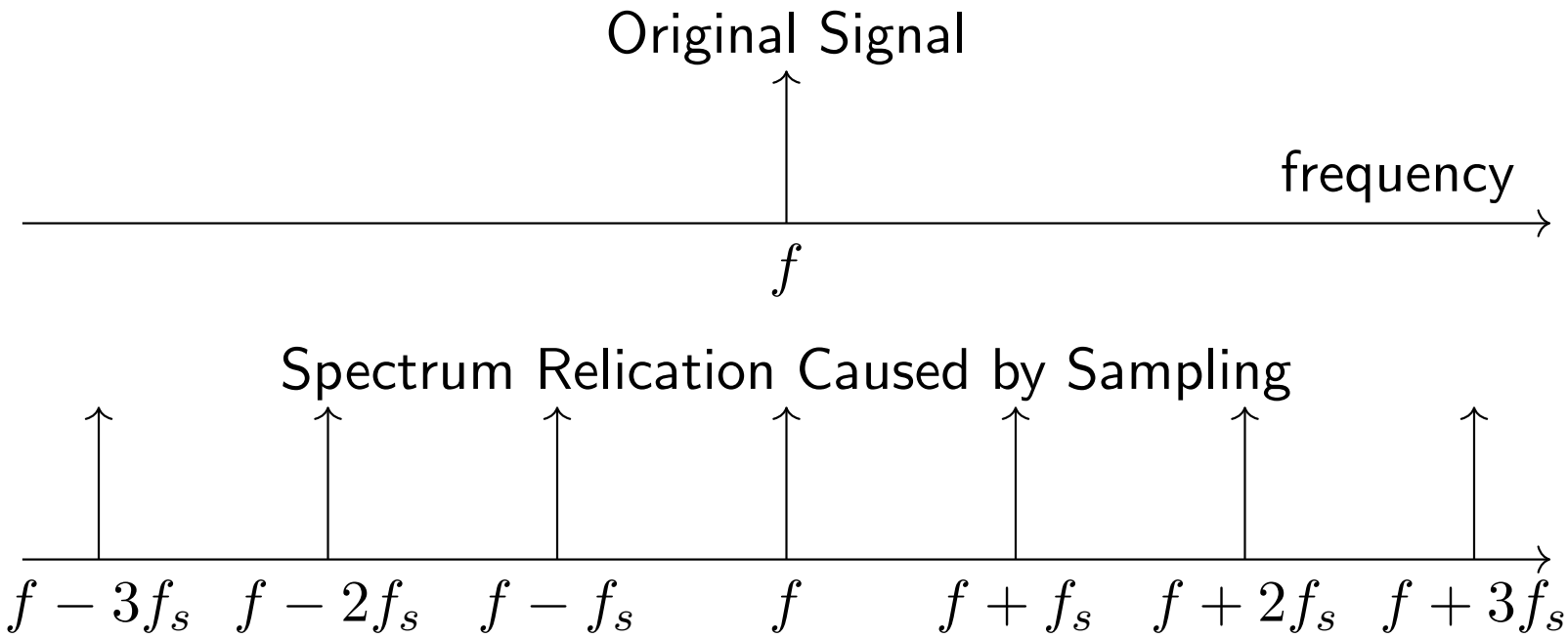
Did the sampled data come from $y(t) = \sin(17t)$?



Sampling: Frequency Domain Effects



Reconstruction of Sampled Signal



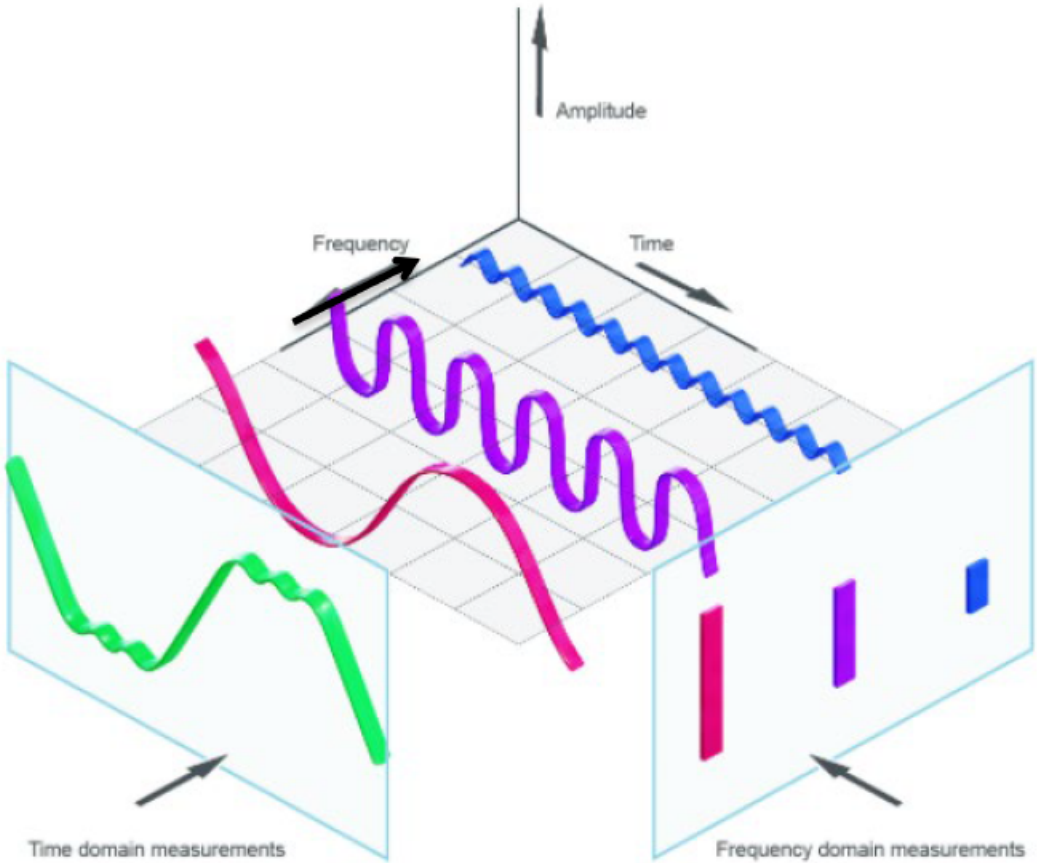
The reconstructed signal contains the portion of the spectrum shown above between

$$\left[-\frac{f_s}{2}, \frac{f_s}{2}\right]$$

Anti-Aliasing

- For time-varying signals if we want to perform any sort of digital signal processing our sampling rate must satisfy the *Nyquist criterion*
- Nyquist criterion: We must sample at greater than *twice* the signal's highest frequency component to prevent aliasing
- The anti-aliasing filter is a low-pass filter that cuts off the signal components that have a frequence greater than or equal to half the ADC sampling rate

Time Domain/Frequency Domain



(Image from https://www.ifm.com/ie/en/shared/technologies/real-time-maintenance/technology/frequency-domain)

- A time-domain signal can be decomposed into frequency components
- The highest frequency determines the *bandwidth*
- Nyquist: must sample the time domain signal at greater than twice the highest frequency component

Conflicting Demands

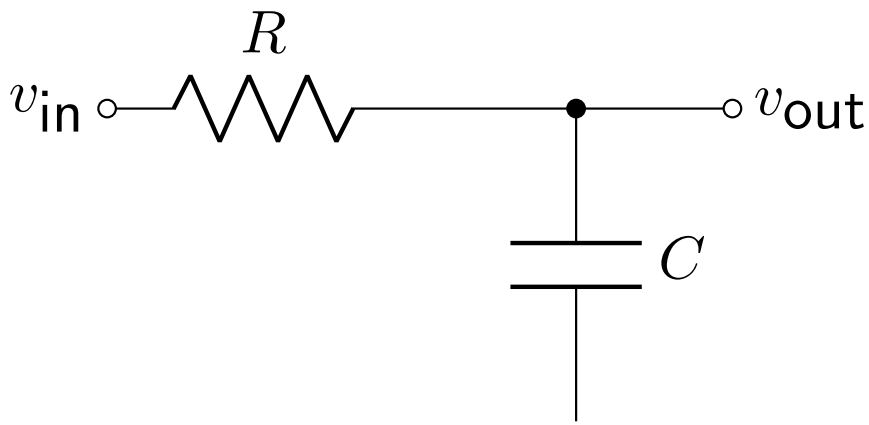
- Nyquist tells us that we need a (relatively) high sampling rate
- A fast-changing signal needs to be sampled more often

- Larger n gives us finer resolution but the A/D conversion takes more time

How Fast to Sample? (Simplistic Model)

- v_{out} is the sampled output
  - We want v_{out} to be very close to v_{in}
  - $v_{\text{out}} = v_{\text{in}}(1 - e^{(-t/RC)})$
  - Error in the sample: $v_{\text{error}} = v_{\text{in}} - v_{\text{out}} = v_{\text{in}} e^{(-t/RC)}$

- When is v_{out} at a useful level for us?
  - When v_{out} is about 1 LSB from v_{in}



How Fast to Sample (Simplistic Model)

- Note 1: $v_{\text{in}} < \text{EFSR}$
- Note 2: $1\ \text{LSB} = \frac{\text{EFSR}}{2^n}$
- Note 3: Want $v_{\text{error}} < \frac{1\ \text{LSB}}{2}$

$$
\begin{aligned}
v_{\text{in}} e^{(-t/RC)} &= v_{\text{error}} < \frac{1\ \text{LSB}}{2} = \frac{\text{EFSR}}{2^{n+1}} \\
e^{(-t/RC)} &< \frac{1}{2^{n+1}} = 2^{-(n+1)} \\
-t/RC &< -(n+1)\ln 2 \\
t &> (n+1)RC\ln 2
\end{aligned}
$$

Effect on Sampling Time

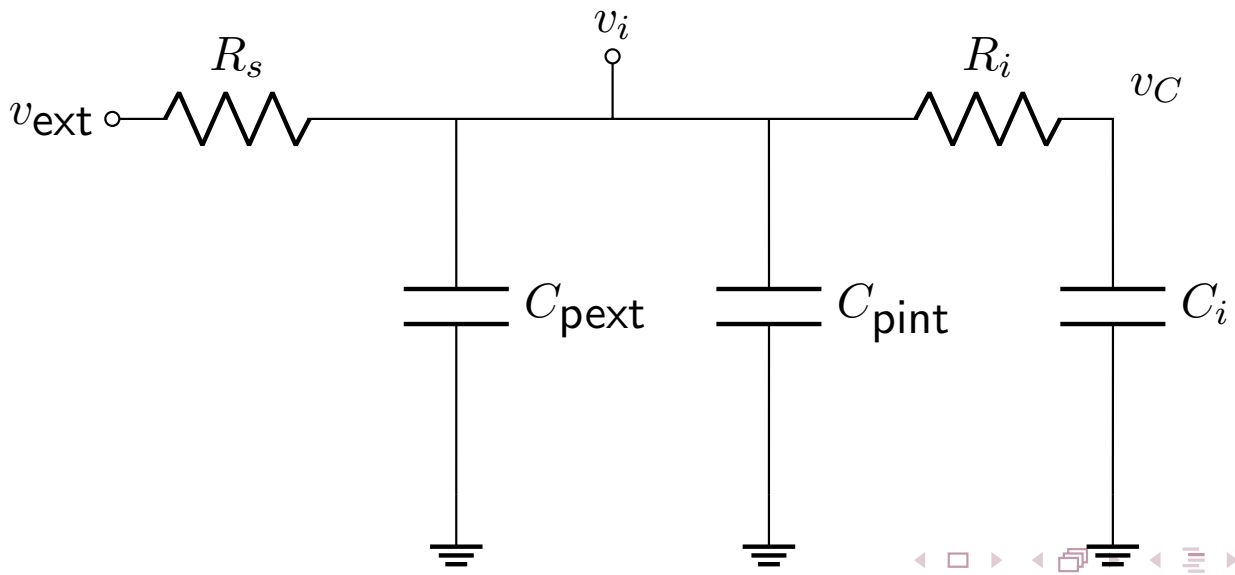
Let t_{sample} be the minimum time to wait for the error between the input signal and its sample to become acceptable.

$$t_{\text{sample}} = (n+1)RC\ln 2$$

- Increasing the desired resolution, increases n, the number of bits and this increases t_{sample}
- Increasing R increases t_{sample}
- Increasing C increases t_{sample}

RC Circuit in the MSP432

- v_i: Input voltage at pin A_x
- v_S: External source voltage
- R_S: External source resistance
- R_i: Internal MUX-on input resistance
- C_i: Input capacitance
- C_{pint}: Parisitic capacitance (internal)
- C_{ext}: Parisitic capacitance (external)
- V_C: Capacitance charging voltage



Sampling Time for MSP432 ADC

$$t_{\text{sample}} \geq (n+1)[(R_S + R_i)C_i + R_S(C_{\text{pext}} + C_{\text{pint}})] \ln 2$$

with

$$R_S < 100\,\text{k}\Omega$$

Example: $R_i = 1\,\text{k}\Omega$, $C_i = 15\,\text{pF}$, $C_{\text{pint}} = 5\,\text{pF}$, $n = 14$, $R_S = 10\,\text{k}\Omega$, $C_{\text{pext}} = 10\,\text{pF}$

$$t_{\text{sample}} = 3.28\,\mu\text{s}$$

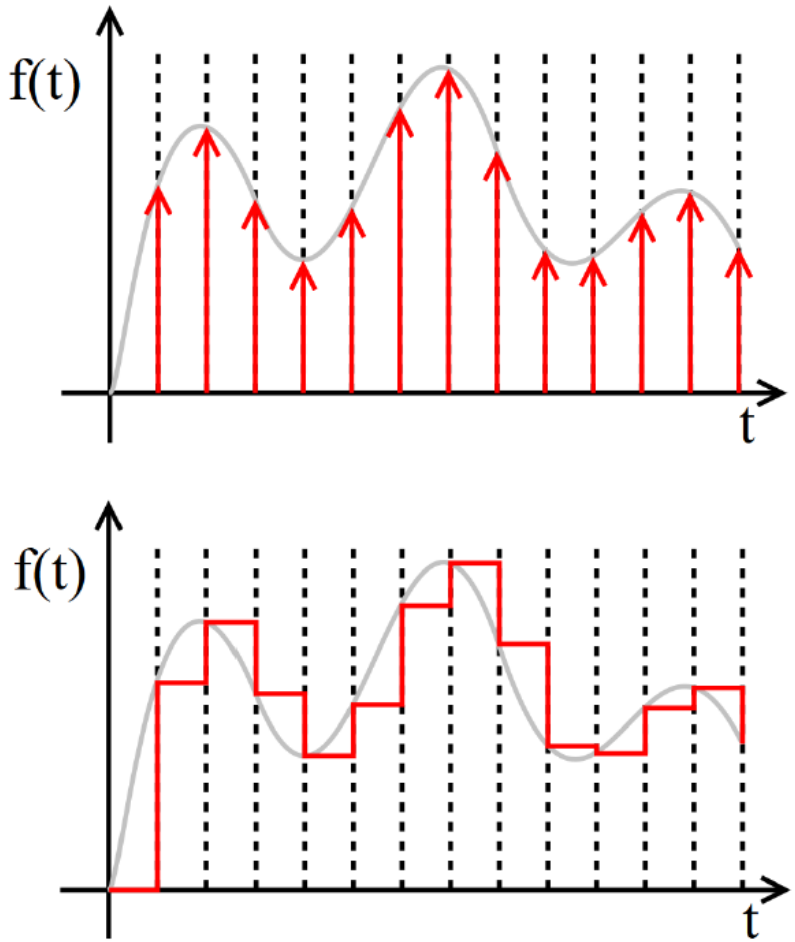
Putting Everything Together

- Let B, for *bandwidth* refer to the highest frequency component of the signal

- Nyquiet rate $> 2B$
  - Sampling period $< \frac{1}{2B}$
- $(n+1)RC \ln 2 < \frac{1}{2B}$

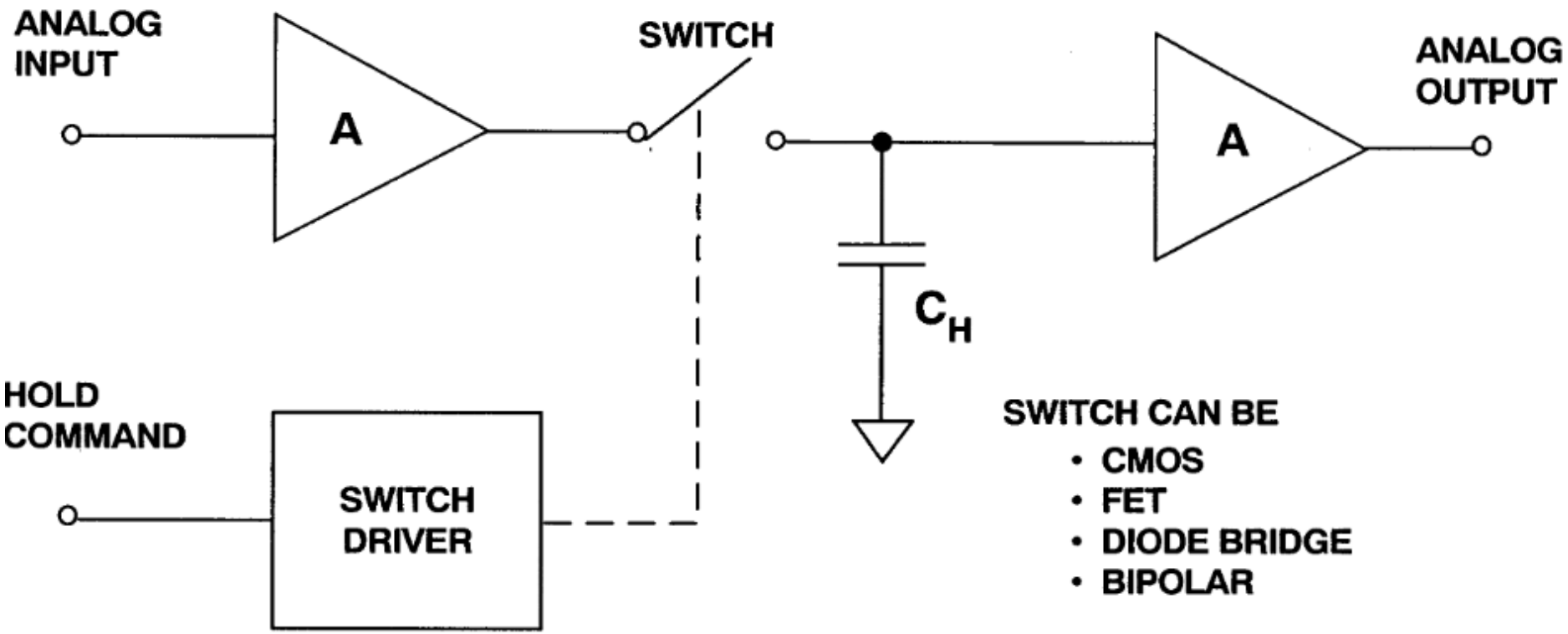
Sample-and-Hold Key Functions

- Purpose: to hold a voltage level steady as and A/D conversion is performed
- Two phases:
  - Sample or "track"
  - Hold or "convert"



Sampling and holding the input signal

Conceptual Sample-and-Hold Circuit [1]

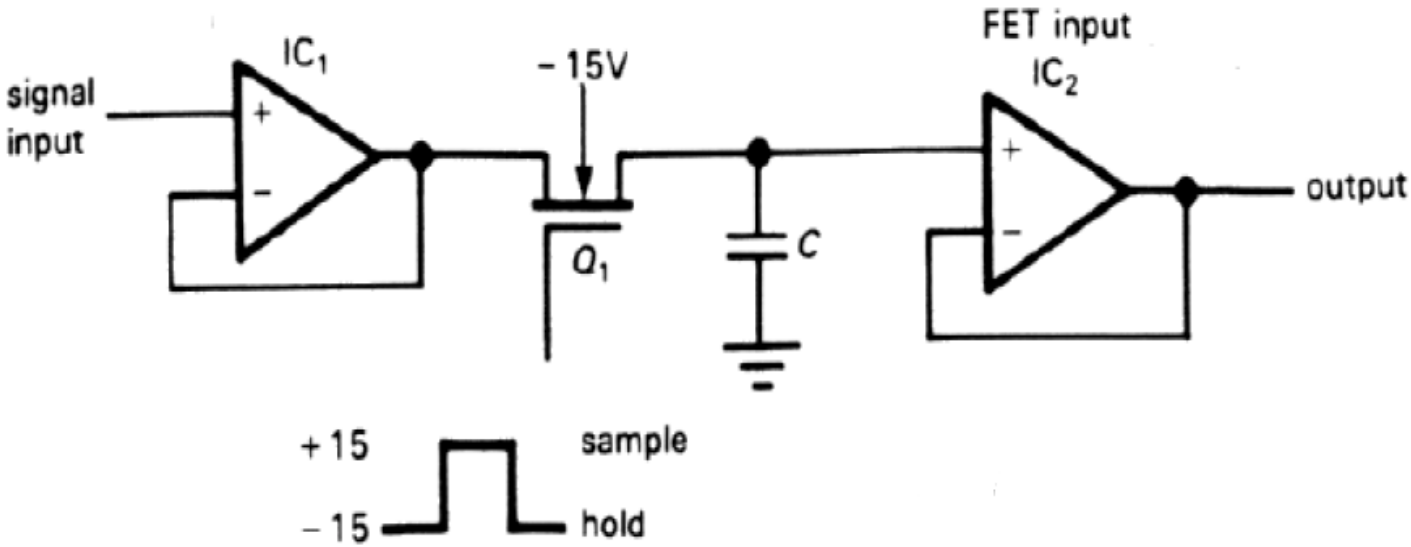


Basic Sample-and-Hold Circuit

[1] Source: https://www.analog.com/media/cn/training-seminars/tutorials/mt-090.pdf

Sample-and-Hold Circuit

- Components
  - FET-based (Field Effect Transistor) switch
  - Capacitor to hold the sampled value



A typical Sample-and-Hold Circuit

Sample-and-Hold Circuit Operation

- Operation
  - The first op-amp provides a low-output impedance version of the input signal
  - The FET passes the signal during the “sample” phase and disconnects during the “hold” phase
  - The capacitor preserves the value during the hold phase
  - The second op-amp is a high input-impedance op-amp that minimizes capacitor discharge during the hold phase